TALES FROM HIGHMERE

Four Roads

Volume One · Sample: Chapters 1–4

Nghia Doan

FOREWORD

A Note from the Scribe

I am a scribe in Highmere. All year round, I write down people's lives and the events of the day, and I stay quietly outside those stories myself.

When the Regent entrusted these documents to me, she said nothing about the manners of the official halls. She simply handed me blank pages. My work is more than writing neat, careful sentences. I must also do right by the real, flesh-and-blood people who are about to step into my pages.

Right now, I do not yet know who the main character will be, or where the story will wander.

Dear reader, let us set the ledgers aside for now. Come and turn the pages with me, one by one.

Nghia Doan

Scribe of Highmere

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PROLOGUE

A Hundred Stones Call for Help

Snow had fallen for three straight days over Highmere. That afternoon, in the highest room of the palace, Regent Orla stood alone before the Great Map.

The map covered the entire wall, showing mountains, rivers, roads, and the name of every town. Beside each town, each harbor, and each mountain pass, a small stone was set into the map. Each stone had a twin in the very place that bore its name, watched over by a stonekeeper. Whenever trouble struck there, the stonekeeper sent word through the twin stone, and the stone on the map would glow—soft as candlelight, but a deep purplish red.

This winter, all one hundred stones on the map were glowing.

No letters had come from those places, and no taxes either. The officials she sent to inspect them brought back only brief reports: the roads were repaired, the gates were open, and all was well. Yet the stones kept glowing. She had read every report with care. She knew which ones lied, she knew who had signed them, and she knew what people had been whispering across the capital since autumn: that a rebellion was coming, and that someone meant to harm the young king before the day he took the throne, on the winter solstice of the coming year.

She sent for the old jeweler, her own uncle, who had served the royal family since the previous king's reign.

"Cut four pieces for me from the stone in the corner of the map," she said. "A necklace pendant, a ring, a pair of earrings, and a bracelet. As soon as they are finished, bring them to me at once. Tell no one."

He bowed his head and turned away without a single question.

At dusk the next day, in the courtyard beside the stables, she met Mira, a young forest ranger who had just returned from the northern mountains with snow still clinging to her shoulders. Orla unrolled a scroll for her to see: the route she would take, with a few short lines written beside each stop. Mira studied it for a moment, then pointed to a mountain pass.

"This road is drawn wrong, my lady. In winter, people have to go around below the pass."

Orla smiled and fastened around Mira's neck the necklace with its pale jade-green stone.

"Near a friend, the stone glows. Near an enemy, each stone warns you in its own way."


Highmere

"Who will I meet?"

"You'll find out. Until then, if you are ever caught, you will have nothing to hide."

Then she instructed her: "At each place on the map, find out what happened, from beginning to end, and whether someone had a hand in it. Find a way to set things right. Then write to me. Quill will find you wherever you are, bring your letters back, and carry my letters to you."

Near midnight, she climbed the observatory tower, where Tavi, apprentice to the court mage, was taking notes beside the telescope. When he heard her tell of the twin stones, he asked at once:

"When two stones are so far apart, how does one know the other is sending a message?"

"Even your master cannot answer that yet," she said. After giving him her instructions, she handed him the ring and a rolled map.

Past midnight, in the Archives, the candles were still burning when she walked in. Suri, the youngest ledger clerk, stood up. Orla handed her a report. Suri read it, then held the paper up close to the candle flame.

"The seal is real. But the ink is far too fresh."

Orla nodded. She had suspected as much. She handed her the pair of earrings and a map of her route.

Before dawn, at the river landing below the city wall, Bao, a young mail boatman, was loading baggage into his boat. When she asked about the places along the river, he answered without hesitation: which landings boats had stopped calling at, and since which month; and which markets had seen the price of salt double. She told him to put on the bracelet and handed him a leather pouch holding his map.

From the tower, she watched Bao's boat drift into the mist. Its lantern shrank and vanished. The other three had already set out in the night, each down a different road, none of them knowing that the others also carried Lady Orla's instructions. Quill the raven landed on the windowsill and shook the snow from her black wings, which shone silver in the early sun.

Behind Orla, on the Great Map, one hundred stones still glowed a deep purplish red.

CHAPTER 1 · MIRA · STONEBRIDGE REST HOUSE

Medicine Across the Bridge ★

From Mira's letters and notes

On the fourth day of winter, a cold wind blew down the rocky streambed from the northern slopes and left a thin frost on the eaves of Stonebridge Rest House. Mira had just leaned her wooden staff by the door and set down her pack when Nela hurried into the courtyard, a travel medicine chest in her arms.

Inside, by the wooden table in the warm main room, the rest house's signal stone glowed a dull purplish red. Ise, who kept the rest house, came out to meet them, one shoulder held low because her arm ached. Nela had not even brushed the snow off her shoulders when Ise spoke:

"Thank goodness you're here. The charcoal burner's boy across the stream has a high fever, and his family has been waiting for medicine since morning."

Mira looked out of the window at the gorge. The big bridge over the stream still lay broken, its span hanging between the gray rock banks. That was why the stone on Ise's table had lit up to call for help. But a stone could not carry medicine across the water.

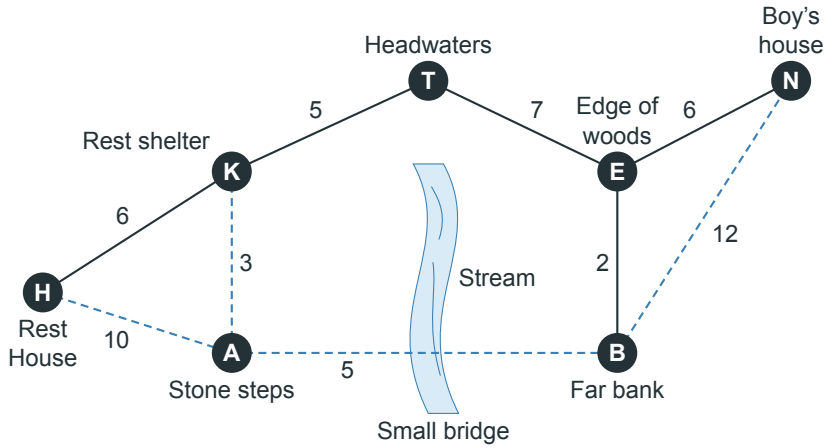
"Let me take it across," Mira said, touching the pale jade-green stone that hung at her neck. She turned to Nela. "Can you still cross the small bridge?"

"You can," Nela said, unlatching the wooden chest. "But the stream has spilled over the path in several places and frozen solid. You have to feel your way, one step at a time, so it's much slower than usual. Still, the way is open."

On the lid of the chest was a map of the trails nearby. Couriers and travelers had written on it how long each stretch really took in these icy days. Across the stream, the boy was still burning with fever. Mira bent over the map. Every minute she lost was a minute he had to wait.

MAIN PUZZLE: WHICH ROUTE IS FASTEST?

The main bridge is broken and cannot be crossed. The diagram below shows seven landmarks and exactly nine path segments that can still be used. The number on each segment is the walking time, in minutes.



Solid lines: ice-free. Dashed lines: icy, but still passable.

Given that:

- Each segment takes the same amount of time in either direction.
- The times shown already include slowing down over icy stretches.
- Walkers can switch from one segment to another only at the lettered landmarks. Apart from the nine segments shown, there are no other links and no shortcuts through the woods.
- The length of a line on the diagram does not show the real walking time.

Questions:

1. Which route should Mira choose to carry the medicine from the Rest House (H) to the boy's house (N) as fast as possible, and how many minutes will that trip take?

2. How can we be sure that there is no faster route?

Hint: A route with fewer segments does not necessarily take less time. Record the shortest total time found so far from H to each landmark, and replace that number with a smaller one whenever you find a faster way to reach that point.

SOLUTION: FINDING THE FASTEST ROUTE STEP BY STEP

Going straight from H to A takes 10 minutes. Going by way of K takes only $6 + 3 = 9$ minutes. We change A's number from 10 to 9 and note that we came from K.[1]

Start with the number 0 at H. On each turn, circle the uncircled point that has the smallest recorded number. From there, add the travel time to each neighboring uncircled point. If you find a faster way, update both the number of minutes and the name of the previous point. Stop once you have circled N.

Circled point	New records from this point (minutes)
H (0)	$K = 6$; $A = 10$.
K (6)	$A : 10 \to 6 + 3 = 9$; $T = 6 + 5 = 11$.
A (9)	$B = 9 + 5 = 14$.
T (11)	$E = 11 + 7 = 18$.
B (14)	$E : 18 \to 14 + 2 = 16$; $N = 14 + 12 = 26$.
E (16)	$N : 26 \to 16 + 6 = 22$.
N (22)	Goal reached; stop.

Trace back from N to H, then read the route forward:

$$H \to K \to A \to B \to E \to N.$$

The total time is $6 + 3 + 5 + 2 + 6 = 22$ minutes.

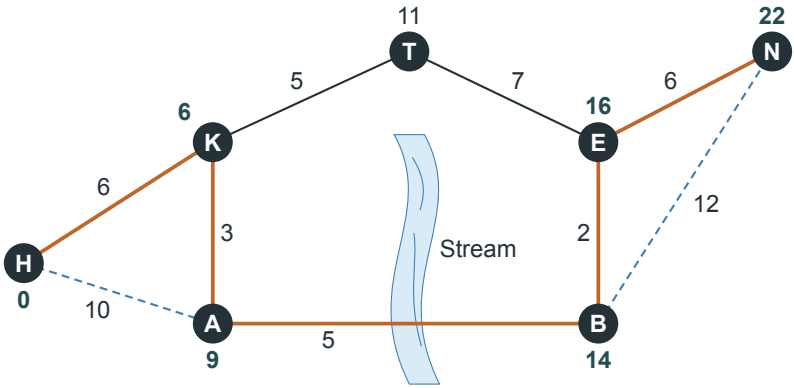
SOLUTION • CHECKING THE CHOSEN ROUTE

The route H $\to$ A $\to$ B $\to$ N looks shorter, with only three segments, but it takes $10 + 5 + 12 = 27$ minutes, 5 minutes slower.

Why can we be sure there is no route faster than 22 minutes?

On each of the nine segments, the two numbers at its ends differ by no more than its walking time. So whichever way you go from H (marked 0) to N (marked 22), the trip takes at least 22 minutes. The route we found takes exactly that long.

[1]The algorithm was published by E. W. Dijkstra in 1959 in Numerische Mathematik, vol. 1, pp. 269–271. https://doi.org/10.1007/BF01386390



Beside each point: its shortest time from H (minutes). Bold line: H → K → A → B → E → N, 22 minutes.

"Up to the rest shelter, down the stone steps, over the small bridge, then along the edge of the woods," Mira said under her breath. She folded the map and took the insulated pouch with the medicine bottle from Nela.

Outside the gate, the mountain wind tugged at the hem of the fourteen-year-old's moss-green cloak. On the icy stone steps, Mira held the pouch tight with one arm and kept her other hand on the rock wall.



At the small bridge, the water still ran, murmuring between two rims of cloudy white ice. Mira did not look down at the stream. She watched the planks and chose where to put each foot. One step, then another. The cold came up through the soles of her boots. The whole way, she kept the top of the pouch shut tight.

Twenty-two minutes after she set out, Mira reached the charcoal burner's porch and put the medicine straight into the hands of the healer who was watching over the boy.



In a warm room that smelled of pine logs, the little boy lay in bed, his cheeks bright red with fever. While the healer got a spoonful of medicine ready, he reached out a hot little hand and held on tight to Mira's finger. When the spoon came near, he scrunched up his face and turned away from the bitter smell.

The healer did not rush him. "Swallow this bitter spoonful, love, and you'll get a sip of sweet warm water right after," she coaxed. The boy squeezed Mira's finger, shut his eyes tight, and swallowed it all.

By the time Mira got back to the rest house, the afternoon sky was turning dark gray. She returned the pouch to Nela and went to wash her hands. When she stepped up to the fire to warm her frozen fingers, she noticed a small creature, about the size of a cat, sitting on the windowsill and gazing out at the darkening sky.

It was a young dragon with coppery-red scales, neat little wings folded close to her sides, and clear amber eyes.

"She got separated from her mother a few days ago," Ise said softly as she came over. "She just sits there all day, looking toward the mountains."

Mira sat down in the chair by the hearth and slowly held out one hand, palm up. The little dragon turned her head. Her amber eyes studied Mira for a long moment. Then she crept forward and touched her warm nose to Mira's fingers.



Soon she had climbed into Mira's lap, curled up like a warm coal, and fallen asleep. Her coppery-red scales gave off a gentle warmth. Mira stroked the tiny ridge of her back and named her Ember, after the glowing coal she looked like.

After supper, when the room was quiet except for the crackle of the fire, a tiny creature crawled out from deep in the ashes. It was no longer than a finger and scurried on tiny legs, and its back glowed a dull red, like a dying ember.

"That's Ashback," Ise said, smiling at Mira's surprise. "It was living in this fireplace long before I came to the rest house. Every evening, when people sit and talk, it crawls out to listen to the stories."

Mira smiled, leaned toward the edge of the hearth, and told how the little boy across the stream had scrunched up his face at the bitter spoonful that afternoon.

Ashback stood still on the edge of the bricks, its soft red back glowing as if it were taking in every word. Only when the story was over did it slip back into the warm ashes and vanish.

That evening, since she was going back to Highmere in the morning, Nela took out a small pouch, tied and sealed, with fourteen coins inside, and gave it into Mira's care:

"When you pass through Birchrest Lodge, please give this pouch to Iven, into his own hands. He's my younger brother, and he looks after the lodge."

Mira wrote the name Iven and the place, Birchrest, in her notebook. Then she put the sealed pouch in its own pocket of her pack, well away from her own money.

Ise was struggling with a pile of bills and medicine delivery slips. Since the bridge broke, the routes had changed, and with her shoulder hurting, the paperwork kept piling up. She hired Mira to put the records in order and to write route directions for the medicine couriers. Mira pulled a chair up to the table.

When the couriers came by, Mira read the directions out to them. One older courier listened, then shook his head:

“The old path through there is blocked off now, child. To reach that house, you have to go around behind the house with the pear tree.”

Mira had never been past the house with the pear tree. She asked him where to turn, and what a stranger should look for in winter, when the pear tree was bare, so as not to knock at the wrong door.

Other couriers back from their rounds added small details: a slick patch of ice hidden in the shade of the trees, a stone step with a chipped edge to avoid. Mira wrote each warning under its place.

Those talks at the wooden table added new notes to her own map, too.

By the fifth day of winter, her pay for the ledgers had covered her room and board and the cost of a letter to Lady Orla.



One evening, Ise dug a pair of iron ice grips out of an old cabinet. They were meant to be tied under boots, but one strap had rotted through long ago.

Her shoulder and arm still ached, so Ise had to rest her arm on the table to sew on a new strap, while Mira sat beside her and held the cord taut so every stitch went in firmly.

When the sewing was done, Ise gave the grips to Mira for the mountain roads ahead.

“They hold well on hard ice,” Ise said, “but don’t wear them on wet planks, or on rock that’s only damp with frost. You’ll slip. And even with grips on, don’t take a chance on every slope.”

Mira thanked her.

On the seventh day of winter, Mira closed the finished delivery ledger, and Ise paid her the rest of her wages.

That afternoon a young courier came back from his rounds, beaming before he was even through the door:

“Those directions are so clear, ma’am! I just followed them and found the house straight away. Not one wrong turn!”

Ise smiled with relief, put the finished ledger on the wooden shelf, and touched the signal stone on the table herself. Its light went out.

On the morning of the eighth day of winter, as Mira shouldered her pack and picked up her staff, Ember left the windowsill and trotted over to her feet.



Mira stopped, bent down, and opened the front of her moss-green cloak. The little dragon climbed in by herself and scrambled up onto Mira's shoulder.

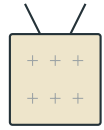
Mira waved goodbye to Ise and stepped out with her new companion onto the winter road, bright in the early morning sun.

YOUR TURN

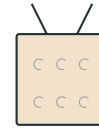
Exercise 1: The Bag That Keeps Medicine Cool ★

Go back to the rest house, just before the 22-minute medicine trip. Earlier, Nela tested two insulated bags with bottles just like the one Mira will carry, holding the same amount of medicine, all starting at $5\,^{\circ}\text{C}$, in the same outdoor weather as today.

The medicine must be kept between $4\,^{\circ}\text{C}$ and $6\,^{\circ}\text{C}$ ($4\,^{\circ}\text{C}$ and $6\,^{\circ}\text{C}$ included). The tests showed:



Linen
26 minutes



Dry wool
32 minutes

During each bag's tested time, the medicine always stayed between $4\,^{\circ}\text{C}$ and $6\,^{\circ}\text{C}$. Beyond that time, nothing guarantees that it stays safe. On the trip, the bag stays tightly closed, just as in the test.

At first, Mira planned to take the medicine bottle out of its $5\,^{\circ}\text{C}$ storage, put it into the bag, and then stand out on the porch for 6 minutes to check her straps and tie her bootlaces before setting off. The temperature on the porch and along the trail is the same.

a) With that plan, which bag should Mira choose for the 22-minute trip, and for at most how many extra minutes could she stop along the way while still staying within the bag's tested time?

b) If only the linen bag is left at that moment, how can Mira change how she prepares so she can deliver the medicine safely without having to walk faster or switch to another route?

Hint: Include the time spent standing out on the porch getting ready.

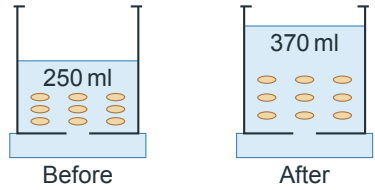
Answer: a) Mira should choose the dry wool bag. If she takes the medicine out before going onto the porch, it spends $6+22=28$ minutes in the bag (porch plus trail), more than the linen bag's 26 minutes. With the dry wool bag (tested for 32 minutes), she can stop along the way for at most $32-28=4$ extra minutes. b) Mira should keep the medicine bottle in its $5\,^{\circ}\text{C}$ storage, finish checking her straps and tying her bootlaces first, and only take the medicine out and place it in the linen bag right before setting off. That way, the medicine is only in the bag for the 22 minutes on the trail, which stays within the tested 26-minute limit of the linen bag.

Exercise 2: Why Did the Soil Rise? ★★

Back from the medicine trip, Mira saw that ice had pushed up the soil in a flowerpot on Ise's porch. Ise had kept the pot's measurements from before the deep freeze.[2]

The earthenware pot is rigid. At first it held 250 ml of wet soil, including 100 ml of water. The hole in its bottom touches an unfrozen water source. After several freezing nights, the soil, water, and ice took up 370 ml.

Both cross-sections have one scale. Blue under the pot: the unfrozen water source; the soil particles are simplified. Water enters only through the bottom hole. No water leaks out and no soil is lost.



The pot has no air gaps, and the solid soil particles keep their volume. Freezing water grows by at most $1/10$ of its volume.

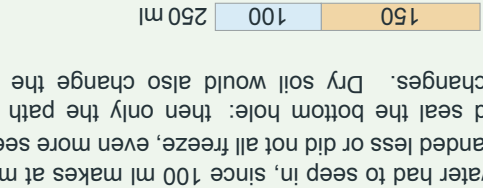
a) Without extra water seeping in, could the volume reach 370 ml? At least how many ml of liquid water seeped in through the bottom hole?

b) To test the bottom hole's role under the same conditions, should Ise use dry soil, or keep the wet soil and seal the hole? Why?

Hint: Find the increase from the first 100 ml of water. Water that seeps in can also freeze.

Answer: a) No. Freezing adds at most 10 ml to the first 100 ml of water, so the pot is still short by at least $370 - (250 + 10) = 110$ ml. To make up those 110 ml, at least 100 ml of extra water had to seep in, since 100 ml makes at most 110 ml of ice. If the water expanded less or did not all freeze, even more seeped in.

b) Keep the wet soil and seal the bottom hole: then only the path for water through the bottom changes. Dry soil would also change the starting amount of water.



Tan: solid soil. Light blue: the first water, or the most ice it can make. Grey: the missing 110 ml of ice, which needs at least 100 ml of water seeping in. Same scale for all bars.

[2] Water from unfrozen ground can seep into soil and freeze into layers of ice, pushing the soil upward. See A. W. Rempel, Formation of ice lenses and frost heave (2007). This problem uses hypothetical data and a simplified model. https://doi.org/10.1029/2006JF000525

CHAPTER 2 · BAO · REEDWATCH LANDING

Mail in Midstream ★★

From the accounts of Bao and Sedge



It was already 8:35 in the morning, and the mail net strung across the water at Reedwatch Landing was still empty.

Bao stood on the pier, one hand holding his conical hat against the river wind, watching upstream. He was only fourteen, but oars and river trips were nothing new to him. In the station house beside the landing, Sedge the heron bent over a table, turning a bamboo mail tube from yesterday's run with the tip of his long beak.

"Look at the wax on this cap," Sedge said, folding his wings against his sides.

Bao went in, tugging at his faded red scarf, and bent to look. Over the old seal, which had been pried apart, lay a fresh, glossy layer of wax. Someone had opened this tube and sealed it again before it floated down to the landing, yet no one had reported a missing letter since yesterday.

What worried them most was that a letter for Lady Orla was due on this route today. Sedge was to catch it and hand it straight to her courier, Bao. But each tube was carved only with the name of its landing; the person's name was on the envelope inside. Anyone hunting for one person's letter would have to open the tubes one by one.

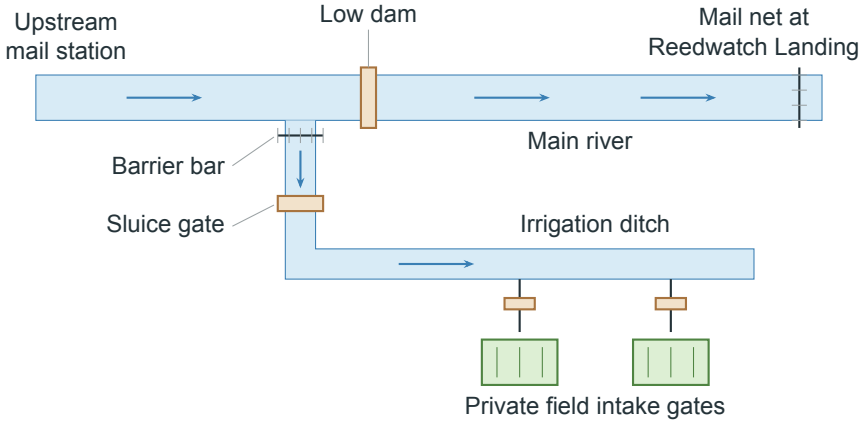
Just then a farmer stomped in, his boots caked with mud.

“Sedge, the water’s been acting strange,” the farmer panted. “Yesterday I opened my gate to water the field, and the water ran for a little while, then stopped dead. This morning it was already coming down the ditch when I got there, much earlier than usual.”

Sedge stopped turning the tube and looked at Bao. Outside, the net was still empty. Bao opened the station ledger and checked the farmer’s story against the mail schedule and the rules for the sluice gate at the low dam above the landing.

MAIN PUZZLE: WHEN DID THE SLUICE GATE OPEN?

Every day, bamboo mail tubes are released into the river from an upstream mail station. They drift through the water in front of a low dam, spill over the top of the dam, and float downstream into the mail net at Reedwatch Landing. Just in front of the dam, an irrigation ditch branches off to carry water to the fields; a barrier bar across the mouth of the ditch keeps the mail tubes from drifting into it. Right behind the barrier bar is the main sluice gate. From this ditch, each household has its own intake gate to let water into its field.



How the water flows (not to scale).

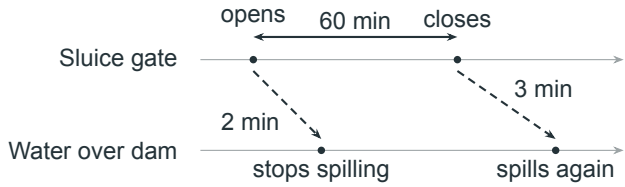
A normal day. The mail tubes leave upstream at 8:00, reach the front of the dam at 8:10–8:14, and arrive at the net at 8:25–8:33. The main sluice gate opens from 10:00 to 11:00. Each household may open its own field gate for as long or as short a time as needed.

When the main sluice gate opens, water turns into the ditch and the water level in front of the dam drops. Any mail tube that arrives while water is not spilling over the dam must wait at the barrier bar. Keep track of each of these separately:

Sluice gate	Opens once, closes exactly 60 minutes after opening.
Water over dam	Stops spilling 2 minutes after the gate opens; spills again 3 minutes after it closes.
Waiting tubes	Once water spills again, each tube takes 15–19 minutes to reach the net.

Yesterday. All eight tubes left the station at 8:00. The first tube reached the landing at 9:22, the last at 9:26. The sluice gate opened only once that day and did not open again at 10:00.

When a mail worker reached the dam, the gate had already closed, but water had not yet started spilling over again. The worker saw all eight tubes in front of the barrier bar and stayed until all of them floated over the dam. During that whole time, no one picked up or held back any tube.



Order of events (not to scale).

Questions:

1. What time did the irrigation sluice gate open yesterday, and what time did it close?

2. From what time to what time did water stop spilling over the dam, and why was that period not exactly one hour?

Hint: Work backward from the time the first mail tube reached the landing, and then from the time the last mail tube reached the landing. What time must water have started spilling over the dam again to satisfy both conditions?

SOLUTION: WORKING BACKWARD FROM ARRIVAL TIMES

We find the time when water started spilling over the dam again yesterday morning by comparing two limits:

- From the first mail tube (arriving at 9:22): Each tube held back at the dam needs at least 15 minutes after water spills again to reach the landing, and 9:22 minus 15 minutes is 9:07. So water must have spilled over the dam again no later than 9:07.
- From the last mail tube (arriving at 9:26): Every tube held back at the dam reaches the landing within at most 19 minutes after water spills again, and 9:26 minus 19 minutes is also 9:07. So water spilled over the dam again no earlier than 9:07 (if it had spilled earlier than 9:07, all 8 tubes would have reached the landing before 9:26).

Combining both limits, we know for certain that water started spilling over the dam again at 9:07 sharp.

From 9:07, work backward using the given delays:

Time	Event	Calculation
8:04	Irrigation sluice gate opens	$9{:}04 - 60$ min
8:06	Water stops spilling over dam	$8{:}04 + 2$ min
9:04	Irrigation sluice gate closes	$9{:}07 - 3$ min
9:07	Water spills over dam again	Both limits above

Thus, the irrigation sluice gate was open for exactly 60 minutes (from 8:04 to 9:04), but the flow over the dam was interrupted from 8:06 to 9:07, which is 61 minutes (or $60 - 2 + 3 = 61$ minutes). This interruption lasted longer than an hour because water took 2 minutes to stop spilling after the gate opened, but 3 minutes to rise enough to spill over the dam again after the gate closed.

“I think someone opened the irrigation sluice on purpose at 8:04 yesterday,” Bao said, looking up. “That held the mail tubes in front of the dam so he could search them, and he left before the station worker came.”

Bao glanced at the clock: 8:40. The dam was twelve minutes upstream. This morning the mail was late and the water had come early. If the same person was playing the same trick, the tubes might still be lying in front of the dam.

“If we go right now, we might still be in time,” Bao said.

“I’m coming with you!” the farmer said at once.

Sedge shut the ledger at once and strode out with them on long legs.

At 8:52 sharp, the three of them crept up to the low dam. River water was pouring through the sluice gate into the irrigation ditch. Behind a woven screen near the barrier bar, a stranger crouched beside a cloth bag full of bamboo mail tubes. Several had their caps off. From where they hid, they watched him pry the wax off a tube with something sharp, pull out the envelope and read the name on it. Then he stuffed the envelope into his own bag.

At that moment, the small pale-pink stone on the thin braided brown cord around Bao’s left wrist turned cold, as cold as river water in winter. It made no sound. But Bao knew: someone who meant harm was close by. He held his breath.

Bao signaled the farmer to circle behind the screen and waved to Sedge to cut off the other end. Sedge tried to tiptoe, lifting his long, spindly legs far too high and setting them down feather-light. Tense as he was, Bao nearly laughed.

Suddenly the stranger spun around. Without a second’s pause, Sedge lunged and clamped his long beak on the mouth of the cloth bag. Behind him, the farmer spread his arms to bar the way out.

In a panic, the stranger yanked the bag toward himself. The cloth ripped, and an envelope flew out, dropped into the ditch and went racing away through the reeds.

Bao dashed to the water’s edge and scooped the letter out just before the current carried it off. But when he looked back toward the dam, the stranger had vanished through the bushes on the bank.

“What a shame, he got away!” The farmer smacked his hands together in frustration.

Sedge still had a torn corner of the bag in his beak. He looked so solemn and so puzzled at once that Bao could not help laughing. He looked down at the envelope he had fished out. The ink on the outside was still clear: “To Lady Orla.”



When he took the scrap of cloth from Sedge’s beak, Bao saw a strange emblem printed on it: three bags tied together with one strap. Three bags, one strap. Bao looked at it a moment longer, then folded the scrap and put it in his pocket.

So the water already in the ditch would not go to waste, the farmer hurried off to tell the neighbors to open their field gates. Bao and Sedge stayed at the dam and gathered up the scattered tubes and the soaked letters.

That afternoon, while the letters dried, Bao looked out at the river and asked:

“Sedge, if every village has to block the river like that to water its fields, how can the mail ever arrive on time?”

Sedge sighed, eyeing the twin stone glowing in the station corner.

“That’s exactly why I lit the stone and called Lady Orla for help, my boy. People’s fields can’t go without water, and carrying it up from the river bucket by bucket takes far too long and is never enough.”

So Bao went with the farmer to look at the field intake. What they needed was an intake gate that would open, give the fields about an hour’s water, and close again by itself, while still leaving water flowing down the main river.

Over the next few days, the people of the area brought timber and lent a hand to build the new intake. The villagers made the guide

chute and the tank lid, and Bao carved the floats, rigged the pull cords and fitted the latch. While he drove wooden stakes into the ditch bank, a villager held the plank steady for the last few blows; down in the ditch, another rolled up his trousers and waded in to clear away the stones that blocked the flow.[1]



During one test, a man from a village downstream walked up along the ditch and stood on the bank, worried:

"If you keep drawing water up here like this, what will happen to the water flowing down to our village?"

Bao put down his tools at once and walked with him all the way downstream along the ditch to see for himself. They came back, adjusted how much water was taken in upstream, and walked down together to check again. Only when the upper fields and the downstream villages both had enough water did everyone take down the old dams that had held the water back.

At the last test, the farmer worked and adjusted the intake gate himself, while Sedge stood beside him keeping time by the clock. Bao stood at the edge of the landing, rinsing wood shavings off his hands in the river. When the water was flowing steadily into the fields and down to the lower villages, Sedge strolled into the station and put out the twin stone himself.

Before Bao left, Sedge gave him the money the villagers had collected to thank him. When Bao offered to pay for his room and board, Sedge waved a wing and would not take a coin.

[1] See the diagram of the outlet gate and latch in Exercise 1.

Bao packed the letter for Lady Orla in a waterproof pouch, laid his oars in the boat and got ready to cast off. Before he could step aboard, Sedge called him back: a timber raftsman had just reported something strange downstream at Whirlpool Basin.

Bao followed Sedge up the high bank and looked down. Out in the middle of the wide water, a long dark shape was gliding under the surface, trailing swirls of white foam before it sank out of sight. Bao stared hard, but he could not tell its head from its tail.



"What is that, Sedge?" Bao asked softly.

"People here call it the Sleeper Beneath the River," Sedge said quietly. "I've lived at this landing all these years and never once seen it."

Bao had meant to row straight across the middle of Whirlpool Basin to save time. Instead he tied his boat to the post again and sat down hard at the edge of the pier, his legs still shaking a little. Soon a timber raft coming down from upstream also pulled in close to the bank to wait.

Only near noon did the dark shape disappear for good and the swirling water calm down. Sedge and the raftsman walked carefully along the bank, checking the familiar channel close by the reeds.

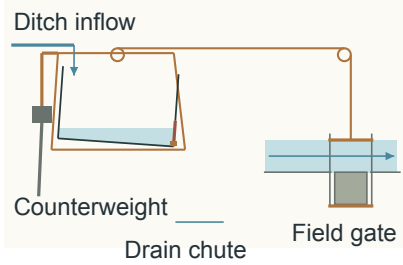
Bao waited a little longer, then let his small boat slide into that channel. He raised his oar high to wave goodbye to Sedge and went on his way, while the flowering reeds disappeared behind the bend.

YOUR TURN

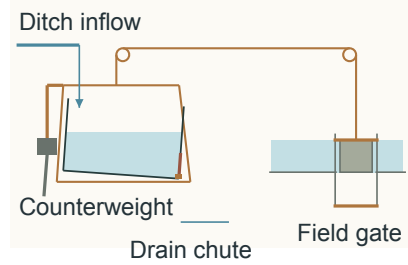
Exercise 1: One More Hour of Watering ★★

At the intake gate beside the ditch, a tank opens and closes the field gate automatically as follows:

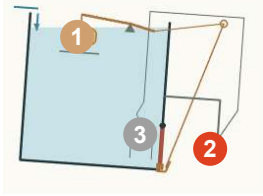
- Water from the ditch flows into the tank at a steady rate of 1 liter per hour. When the tank holds 24 liters of water, the upper float (1) rises and pulls a cord to release the latch.
- The tank's outlet gate swings on a hinge along its top edge; when it swings open to drain the water, the hook (2) holds the outlet gate open.
- Near the end of draining, the water level in the tank drops, lowering the lower float (3) and releasing the hook (2). By then the outflow is weak enough that the outlet gate drops shut under its own weight, and the latch catches again as before.
- Once the water drains out, the lighter tank is pulled up by a counterweight. The field gate lowers, letting water flow into the field. After the tank collects 1 liter of new water, it grows heavy enough to raise and close the field gate.



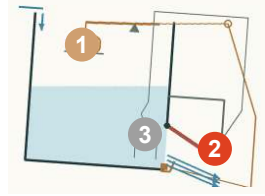
Light tank: gate lowers, ditch opens



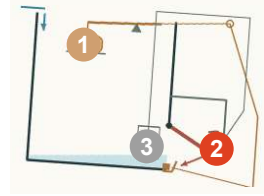
Heavy tank: gate rises, ditch closes



24 liters full: latch releases



Draining: hook holds open



Nearly empty: hook releases

(1) Upper float; (2) hook holding outlet gate; (3) lower float. The tank may begin rising while the outlet gate is still open. Cords, latch, and hook move with the tank.

Now the villagers want each watering to last about 2 hours, while keeping the time from the start of one watering to the start of the next at about 24 hours. Because the tank's outlet is wide, draining the tank and moving the gates takes at most 2 minutes in total, which can be ignored in this approximation. Sedge suggests two plans:

- Plan A: Reduce the rate of water flowing into the tank to 0.5 liters/hour.
- Plan B: Keep the inflow rate at 1 liter per hour, but adjust the counterweight so the field gate closes only after the tank collects 2 liters of new water (the water levels at the start and end of draining stay the same).

Questions:

Which plan meets both of the villagers' requirements? What would happen under the other plan?

Hint: Consider both how long water flows into the field during each watering and the waiting time between two consecutive waterings.

Answer: Plan B meets both requirements.

- With either plan, each watering lasts about 2 hours (Plan A: 1 liter at 0.5 liters/hour takes 2 hours; Plan B: 2 liters at 1 liter/hour also takes 2 hours).
- But with Plan A, water enters the tank at only 0.5 liters per hour, so filling it to 24 liters for a new cycle takes about 48 hours, so the waterings come twice as far apart. With Plan B the inflow stays at 1 liter per hour, so waterings still come about every 24 hours.

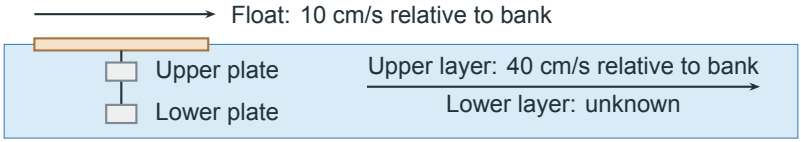
(Note: this is a simplified model. A real device must be tested to make sure the outlet gate drops shut and the latch catches again after every draining.)

Exercise 2: The Float That Drifts Upstream ★★★

To study the current near the surface and deeper down, Sedge puts a float in the river. It sits partly underwater, with two identical metal plates below it: one in the upper layer of water, one in the deeper layer.

In its initial steady motion:

- The float drifts downstream at 10 cm/s relative to the bank: the same way as the upper layer, but slower.
- The upper layer flows downstream at 40 cm/s relative to the bank. We want the velocity of the lower layer.



Given that:

- Each layer pulls its plate the way it moves relative to the float.

- The pull on each plate is proportional to the square of the water's speed relative to the float (if that speed doubles, the pull becomes four times as strong).[2]
- In this steady motion, the two pulls are equal and opposite.

Questions:

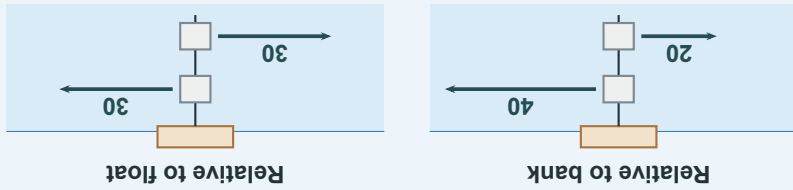
a) Which way does the lower layer flow, and how fast, relative to the bank?

b) Now replace the lower plate with several identical plates side by side at the same depth, none blocking the flow to another (their pulls add up); both layers keep their direction and speed. If we hold the float still relative to the bank and let go without pushing, how many lower plates, at least, make it start drifting upstream?

Hint: When the float is held still relative to the bank, how does the speed of each water layer relative to the float change?

Answer: a) Relative to the float, the upper layer flows downstream at $40 - 10 = 30$ cm/s. Since the two identical plates experience equal and opposite forces, the lower layer must flow upstream relative to the float at 30 cm/s. Because the float itself drifts downstream at 10 cm/s relative to the bank, the lower layer flows upstream relative to the bank at $30 - 10 = 20$ cm/s.

b) When the float is held still relative to the bank, the water speeds relative to the float in the upper and lower layers are 40 and 20 cm/s. Since force is proportional to the square of speed, the downstream pull on the single upper plate is $(40/20)^2 = 4$ times the upstream pull on each lower plate. Four plates only balance the forces; at least **5 plates** are needed in the lower layer for the float to start drifting upstream.



Free-drifting float. Water speeds (cm/s); downstream is to the right.

[2] In practice, the float's motion is also affected by wind, drag on the float body, and tilting. See NASA Glenn: The Drag Equation.

CHAPTER 3 · TAVI · ANVILFIELD FORGE

Fitting Spokes to the Rim ★★★

From measurements and stories at the forge



Out in the yard of Anvilfield Forge, a freight wagon sat propped on wooden trestles, one wheel missing. Inside the workshop, Berrit had finished bending and joining the new rim: round, and true to the millimeter.

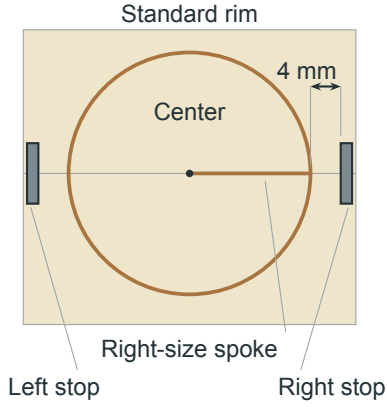
Tavi carried the eight freshly turned spokes, already joined at the hub, over to the inspection table. He was eager to measure, trim and fit the wheel with his own hands. The thirteen-year-old had taken off his rust-orange cloak and rolled up the sleeves of his slate-blue work shirt, and his sandy-blond hair was still a mess. The honey-colored stone in the ring on his right hand glowed softly in the firelight, though today's job needed no magic at all.

Berrit the bear stood by the table and quietly watched. Tavi set the spokes onto the center pin, between the two stop blocks at the ends of the measuring table. A curl of wood shaving had caught under the ruler; he brushed it away before laying the ruler down to read it. Looking at the rough spoke ends, he thought: a wrong number on paper can always be crossed out. But once a spoke is sawed too short, there is no putting the wood back.

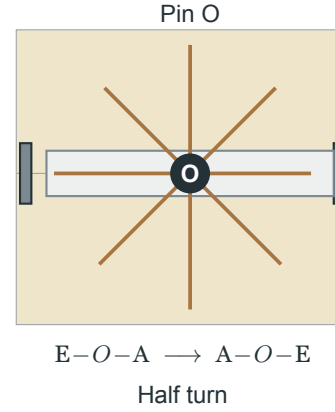
MAIN PUZZLE: IS THE CENTER PIN OFF-CENTER?

Berrit's rim is perfectly round and of standard size. Eight spokes, A to H, form one rigid assembly, spaced evenly around the hub, 45 degrees apart. Opposite pairs across the hub: A and E, B and F, C and G, D and H.

Using the standard rim, Berrit sets two fixed stop blocks on the table, left and right, equally far from the true center, then lifts the rim away. During measurement, the spoke assembly turns around the center pin O on the table.



1. Set both stops
against the checked rim.



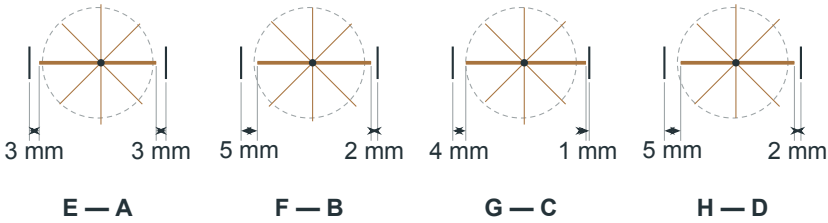
2. Lift the rim away
and measure; pin O stays put.

Reading the gaps. Both stop blocks sit outside the standard rim. When the pin is at the true center, a spoke of the right length touches the rim and leaves a 4 mm gap to the stop block. A 3 mm gap means the spoke is 1 mm too long; a 5 mm gap means it is 1 mm too short. Those numbers apply only on the measuring table. When fitted into the real rim, the spoke ends must sit flush against the rim.

What turns? The two stop blocks stay fixed. The pin can slide left or right, but during measurement it is locked in place, even when remeasuring. Only the assembly of eight spokes turns around the pin. After a half turn, the spoke pointing right points left. We do not yet know whether the pin is at the true center or off-center.

Spokes can only be cut shorter. Cutting 1 mm off a spoke end widens its gap by 1 mm; a spoke that is too short cannot be lengthened.

In the first round of measurements, turning each pair of spokes in turn to face the two stop blocks, Tavi records the following gaps:



Dashed circle shows the unmounted standard rim; diagram not to scale.

Seeing that the opposite spokes E and A both have an equal 3 mm gap on either side, Tavi quickly jots a note on his measurement sheet and suggests cutting 1 mm off both E and A so that each reaches 4 mm. Before the saw blade can touch the wood, Berrit sets her bear paw on the edge of the table: “Hold on.”

Questions:

1. Are the initial measurements enough to prove that the center pin is off-center?

2. If all eight spokes must be kept (only trimming them, without discarding any spoke), which way must the center pin be off-center, and by how many millimeters from the true center?

3. Which spoke would you turn to which side for just one extra measurement to test that possibility, and what measurement result would confirm it?

4. If the test confirms it, how should the center pin be moved, and which spokes must be trimmed, by how many millimeters each?



Hint: Moving the center pin to the left makes the gaps on the left smaller and the gaps on the right larger. By at least how many millimeters must you move the pin to the left to save spokes F and H? And by at most how many millimeters can you move it to the left before spoke A becomes too short?

SOLUTION: SPOKE LENGTH OR PIN OFFSET?

1. We cannot yet conclude that the pin is off-center

The initial measurements could still happen with the pin at the true center, if spokes F and H were already 1 mm too short.

2. To keep all eight spokes

Let d (mm) be the distance the pin must move left to reach the true center. Since spokes can only be cut shorter, every gap after centering must be at most 4 mm:

Spoke	Gap after moving	Condition
F, H (left)	$5 - d$	$5 - d \leq 4$, so $d \geq 1$.
A (right)	$3 + d$	$3 + d \leq 4$, so $d \leq 1$.

Together, the two conditions force $d = 1$: the pin must be offset to the right by 1 mm. This is still a hypothesis needed to save all eight spokes, and it must be tested before cutting.

3. Extra test measurement

To test this, keep the center pin in place, turn any one spoke to the opposite side, and remeasure its gap.

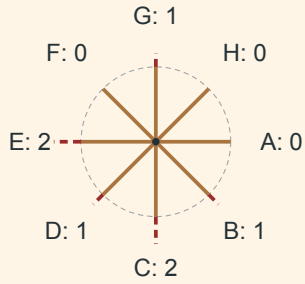
Take spoke A as an example: initially, when A is on the right, its gap is 3 mm. Keeping the center pin locked, turn the spoke assembly half a turn to bring spoke A to the left stop block. If spoke A's new gap on the left measures 5 mm, then the pin's offset to the right is indeed $(5 - 3)/2 = 1$ mm.

Only if A's gap on the left measures 5 mm do we move the center pin back to the true center and start trimming; if the result is not 5 mm, we cannot save all eight spokes only by cutting them.

4. Centering the pin and trimming the spokes

Once the measurement confirms the 5 mm gap, move the pin 1 mm to the left. Left gaps shrink by 1 mm and right gaps grow by 1 mm; then trim each spoke to a 4 mm gap:

Spoke	A	B	C	D	E	F	G	H
New gap (mm)	4	3	2	3	2	4	3	4
Trim off (mm)	0	1	2	1	2	0	1	0



Red dashed end: trim off (mm).

The two initial 3 mm gaps at A and E were equal only because the pin was off-center; they did not prove the two spokes were equal in length. Trimming 1 mm off both would leave A 1 mm too short and E still 1 mm too long. After centering, A needs no trimming and E needs 2 mm trimmed off.[1]

Without moving the center pin, Tavi carefully turned the spokes half a turn, bringing the end of spoke A to the left stop block, and measured again. The ruler read exactly 5 millimeters.



Berrit nodded. She loosened the lock, slid the center pin 1 millimeter to the left and locked it again. Tavi crossed out his hasty calculation and trimmed each spoke to the new numbers. Each time he finished one, Berrit had him turn it to the stop block and measure it. Three spokes, A, F and H, never touched the saw at all.

When every gap on the measuring table was exact, Berrit lifted the spokes to test them in the rim. Tavi stood to the right of the table and held the wooden hub steady so nothing tilted while Berrit guided each spoke end into its socket.

[1]The rim used to set the two stop blocks on the measuring table was checked against the standard size beforehand. In measurement, if the reference standard itself is off, the parts produced may match one another yet all be the wrong size.

When the rim sat snugly around all eight ends, Tavi bent down to look at spoke A, the very spoke he had been about to trim by 1 millimeter only minutes before. Its untouched end now sat firm and solid in the rim.

Tavi let out a long breath, looked up at Berrit and scratched his head.

“Good thing your words were faster than my hand on the saw!”

Berrit laughed, and her deep voice rumbled around the corner of the workshop.

Just then an apprentice came over and picked up Tavi’s notes. “Why is there a crossed-out calculation in the corner here?”

Tavi went a little red. Then he took the sheet, smoothed it flat on the table and pointed straight at the crossed-out line.

“When I saw a 3-millimeter gap on both sides, I thought those two spokes were the same length, and I almost cut them both. Good thing Berrit stopped me in time.”

The apprentice did not laugh. He pulled up a chair beside Tavi and ran a finger from spoke A’s reading on the right to its reading on the left. Then he picked up a piece of charcoal and drew an arrow joining the two numbers.

On a clean sheet, Tavi neatly wrote out the steps for centering the pin and measuring the spokes. The apprentice pinned it to the workshop wall and drew the charcoal arrow again, right beside the step where you turn the spokes.

On the day he left, Tavi collected his pay for the few days he had helped in the workshop. Then he took out his first sheet of measurements, smoothed its curled corner, folded it and tucked it into his leather shoulder bag.

The crossed-out calculation was still there.



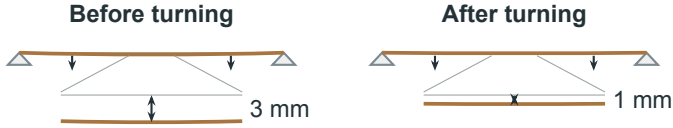
YOUR TURN

Exercise 1: How Bent Was the Bar Already? ★★

While talking with the apprentices in the workshop, Berrit told of a time she nearly straightened a round iron bar the wrong way, and how glad she was that she remembered to turn the bar over and check it before striking with her hammer.

A round iron bar rests horizontally on two supports at its ends. The straight line joining the centers of the two ends serves as the reference line. Initially, the midpoint of the bar sits 3 mm below the reference line.

Next, the smith turns the bar half a turn around its own lengthwise axis (the two ends do not swap places and remain at the exact same height). When remeasured, the midpoint of the bar sits 1 mm below the reference line.



Close-ups of the middle, same scale. Down arrows: weight.

Given that:

- The iron bar itself has a built-in bend; in the initial position, this built-in bend curves downward, and after a half turn around the lengthwise axis, the built-in bend curves upward.
- In both measurements, gravity (the bar's own weight) pulls the bar downward. In this problem's simplified model, the downward sag from weight at the midpoint stays the same when the bar is turned.[2]

Questions:

a) How many millimeters are the bar's built-in bend and the downward sag from weight at the midpoint, respectively?

b) Why shouldn't Berrit straighten out the full 3 mm seen in the first measurement?



Hint: Draw one arrow for the bar's built-in bend and another arrow for the sag from weight; after the bar is turned half a turn, which arrow reverses direction?

[2] A half turn separates the bar's built-in bend (which reverses direction when turned) from gravity sag, which always pulls downward. In illustrations, the bend is exaggerated so the eye can see it easily. In practice, a bar's sag can change after turning; this problem assumes constant sag to make the testing principle clear.

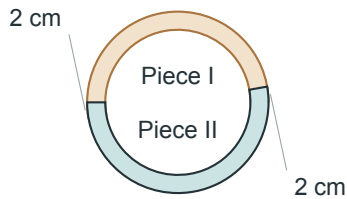
Answer: a) Let the built-in bend be c (mm) and the sag from weight be w (mm). Weight always pulls downward; the built-in bend reverses direction when turned: Before turning: $w + c = 3$; after turning: $w - c = 1$. Adding the two equations gives $2w = 4$, so $w = 2$ mm and $c = 1$ mm. b) Only the 1 mm built-in bend needs to be straightened out. Straightening the full 3 mm from the first measurement would bend the bar itself 2 mm the other way.

Exercise 2: Using Up All the Leather Strips ★★

The workshop apprentices are given four leather strips, each 100 cm long, to make three closed belt loops, each with a circumference of 128 cm. They must use up all the leather given. Instead of picking up a knife right away, the group takes out paper to sketch and calculate how to divide the strips first.

Given that:

- At every joint (including the final joint that closes a belt loop), the two leather ends must overlap by 2 cm. Each overlapped joint reduces the loop's circumference by 2 cm (not 4 cm) compared with the total length of the separate leather pieces. A belt loop made from two pieces of leather has two joints.
- The leather is thin and neither stretches nor shrinks; cutting wastes no length.
- Each piece of leather used must be at least 10 cm long.
- Each cut divides a single piece of leather into two smaller pieces; you may not fold a strip or stack multiple pieces to cut them at the same time.



Joined from two pieces. Example has two joints; each joint overlaps by 2 cm.

Questions:

a) At least how many cuts are needed to make all three belt loops as required?

b) Draw a diagram showing one way to cut and join all three belt loops.

Hint: By how many centimeters is the total circumference of the three belt loops shorter than the total initial length of the four leather strips? How many joints does a closed belt loop made from two pieces of leather have?

Answer: a) Four leather strips have a total length of $4 \times 100 = 400$ cm; three loops need $3 \times 128 = 384$ cm. The difference, $400 - 384 = 16$ cm, goes into the 2 cm overlapped joints, so there are $16/2 = 8$ joints. In closed loops, the number of joints equals the number of leather pieces: 8 pieces are needed. Each cut increases the number of pieces by one, so starting from 4 strips takes at least $8 - 4 = 4$ cuts.

b) One way: keep three 100 cm strips whole; cut the fourth strip into one 32 cm piece and four 17 cm pieces. The total is $32 + 4 \times 17 = 100$ cm, and every piece is at least 10 cm long. Join them as shown: the first loop has two joints (losing 4 cm); each of the other two loops has three joints (losing 6 cm). All three loops have a circumference of 128 cm, use up all the leather, and take only 4 cuts.

One overlapped joint

The 2 cm overlap is counted twice when adding the two strips, but only once in the joined length. It does not lose 4 cm.

$132 - 4 = 128$ $134 - 6 = 128$ $134 - 6 = 128$

CHAPTER 4 · SURI · BRONZEPLATE BOATYARD

One Ring Each Year ★

From Suri's letters and the boatyard files

Wind from the landing gusted under the porch roof of Bronzeplate Boatyard, smelling of fresh mud and damp sawdust. A woman who carried vegetables to market set her rolled-up sail down beside two empty baskets, held her scarf tight against the cold and peered into the yard. Her family's boat had been stuck here for days, and they had been catching rides on other people's boats to get their vegetables to market on time.

Inside the big shed, three wooden boats sat still on their cradles. Maret, the otter who owned the boatyard, had not let any of them be launched yet. She smoothed her whiskers with a front paw, her long tail thumping softly on the plank floor. "Someone mixed up the boats' files," she said. "Now I can't be sure which patches have been inspected and which are still waiting." Beside the doorframe, the boatyard's signal stone was still glowing purplish red.

Suri took the stack of papers from the apprentice and squared it up on the wooden table. The fifteen-year-old pushed her glasses up her nose, and her pale-blue stone earrings swung beside her cheeks as she bent to open each folder.

Three boats were being repaired in this batch: the Swallow, the Willow Leaf and the Silver Wave. The hull in front of Suri and Maret was one of those three, but they did not yet know which. Each boat's file clearly recorded the year it was first launched.

Leafing through the jumbled papers, Suri found an inspection slip dated year 416. The sketch of the boat and the marks noted on the slip matched the hull on the cradle in front of them. So this boat had already been built and launched before the inspection at the landing in year 416.

Maret went to the side of the boat and showed Suri a rib timber that had been patched. Part of it was wood A, from when the boat was first built; the other part was a piece B fitted in during a later re-

pair. Then Maret opened a desk drawer and took out the boatyard's master chart of yearly tree rings. Suri leaned close to the grain, read each ring from the pith out toward the bark, and compared the pattern with the master chart.



MAIN PUZZLE: WHAT IS THIS BOAT'S NAME?

The boat owner is waiting to take her vegetables to market. The boatyard must find out which name in these three files belongs to the boat on the cradle. Below each name is the launch year.

The sketch and distinguishing marks on the year 416 inspection slip match the hull. By then, the boat had already been launched.

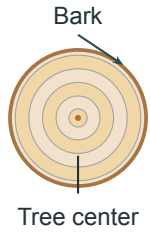
Two wood samples on the same rib timber come from the same local tree species. **A** is the original wood. Its outer edge was trimmed, so rings near the bark may have been lost. **B** is a patch fitted in later, with its bark-side edge intact: the outermost ring of B is the last ring the tree formed.



Swallow
408

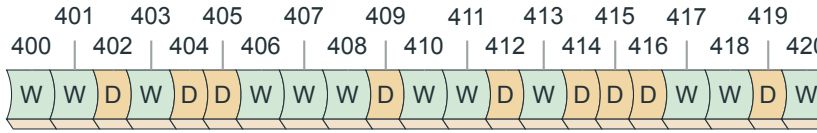
Willow Leaf
414

Silver Wave
418

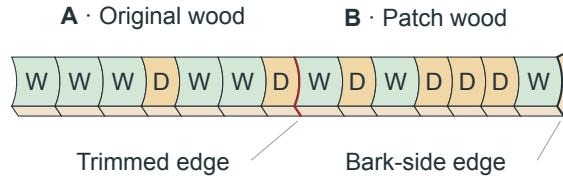


Each year, a tree adds one ring under its bark around the older wood. A felled tree grows no new rings.

Read from the center outward to the bark, from older years to newer years. For the tree species and region here: W is a wet year, when the ring is usually wide; D is a dry year, when the ring is usually narrow.[1]



Maret's boatyard master tree-ring chart · years 400–420



Two wood samples on the same rib timber of the boat

Questions:

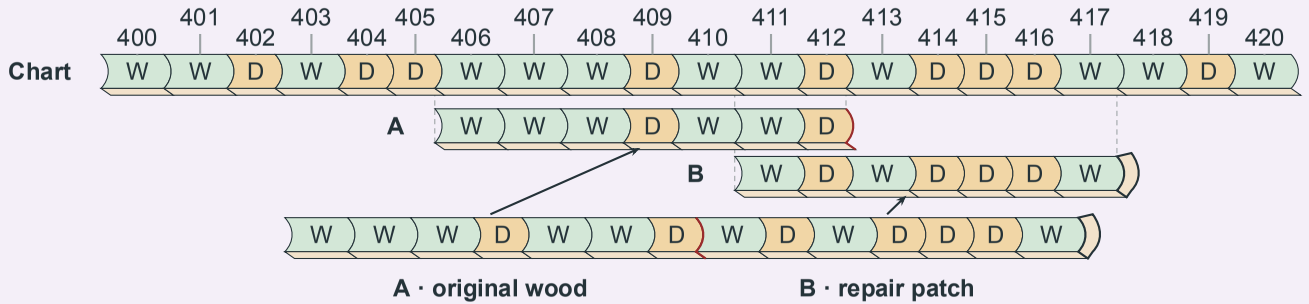
1. Match the ring sequences of sample A and sample B against the boatyard's master chart to choose the right file for this boat. Why do the other two files not fit?

2. Which wood sample still keeps the final ring formed before the tree stopped growing new rings? Why can't the year of that ring be taken as the year the boat was built?

Hint: Consider the original rib timber, the later wooden patch, and the year 416 inspection slip separately. Each piece of evidence sets a different time limit for the boat.

[1]The master chart in this problem is built from local wood samples of the same species and weather logs; the sequences are excerpts from longer verified series. Each symbol represents one year, with no missing or false rings. Ring widths are illustrative, not measurements. In practice, tree-ring dating requires matching many rings across many samples; width also depends on tree species and growing conditions, so the wet/dry rule does not apply to every tree or region. See: University of Arizona, `https://www.ltrr.arizona.edu/eg/skelsurvey.html` ; Historic England, `https://historicengland.org.uk/images-books/publications/dendrochronology-guidelines/`

SOLUTION: MATCHING TREE RINGS AND TIMELINES



Choosing the boat file. A matches years 406–412; B matches years 411–417.[2]

A still has the ring for year 412: the tree could not have been felled before that year. This rules out Swallow, launched in 408.

The year 416 inspection slip confirms the boat was launched before then. This rules out Silver Wave, launched in 418. Only **Willow Leaf, launched in 414**, remains.

Outermost tree ring. B still has its bark-side edge: 417 is the last year the tree formed a ring. B is a repair patch, not the original wood used to build the boat. Because timber can be stored before use, the exact year of the repair is unknown.

A was trimmed: the 412 ring is only the last ring remaining. Outer rings may have been shaved away, so we only know the tree was not felled before 412.

[2]Finding a substring is also a problem in bioinformatics. For DNA sequences (A, C, G, and T), alignment also accounts for mismatches, insertions, and deletions. See: NHGRI, `https://www.genome.gov/25020002/online-education-kit-bioinformatics-finding-functions`

Suri set the Willow Leaf's file beside the cradle. Maret read the inspection slip once more and chalked the unfinished repairs on the hull.

The boat owner swung the rolled sail back onto her shoulder. Her boat had its name again, but it still could not carry her vegetables to market today. Maret told her to come back in a few days, once the joints had been inspected and strengthened.

On the agreed day, when the boat owner came back to the landing, the apprentice carried the nameplate out of the workshop. Maret fixed it to the side of the boat herself and asked Suri to step back and check that it was clear. In the bright sun over the yard, the words Willow Leaf stood out sharp and clean. Suri filed the matching slip in the archive folder, then ran out to help the workers lower the boat into the water.

Once the boat was tied up against the pier, its owner stepped down into it with the rolled sail in her arms, cheerfully showing everyone where her two baskets usually went. She bent and ran a rough hand over the smooth new wood of the patch, and Maret showed her each spot that had been repaired.



Not wanting to miss another market day, the boat owner pulled the sail tight, untied the rope and pushed off. Suri stepped back onto the pier.

From the boat, the owner raised a hand to wave goodbye. Small waves slapped the hull and rocked the boat, tipping one of the baskets; holding the sail line steady, she nudged the basket upright again with the toe of her foot.

Behind Suri, Maret stepped inside the workshop door and turned off the signal stone. Far away, in the high room at Highmere, the matching stone on the Great Map quietly stopped glowing.

Suri cleared a corner of the table and wrote a letter to Lady Orla. By the time she put down her pen, Maret had already pulled the stack of unchecked files toward herself; the boatyard crew would handle the rest. Suri asked her:

“Is there a way to travel on without hiring a cart or an animal to ride?” Maret pointed to the open ground at the end of the riverbank, where Leth was getting a hot-air balloon ready to fly to the landing field near Splitroad House.

When Suri asked, Leth happily agreed. “Hop in the basket and come along. After we land, just help the crew fold the balloon. We have to wait until the fabric is completely cool before we fold it and pack it away.”

The balloon needed a good wind, so they had to wait, and if a storm came, the flight might be put off.



Only in the morning, when a good wind was blowing steadily, did Suri climb into the wicker basket. She set her small bag at her feet and gripped the rim with both hands. Leth made one last check and let the balloon rise slowly from the ground.

The tiled roofs sank lower and lower. Out at the landing, the tops of the tall masts were now level with Suri's eyes. She looked up at the open sky ahead.

YOUR TURN

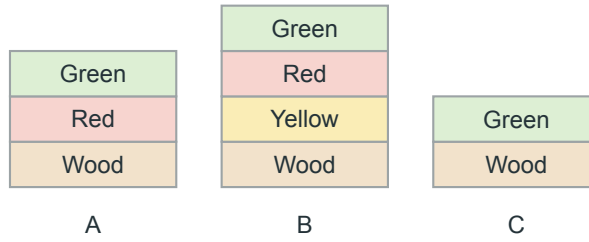
Exercise 1: Under the Green Paint ★

Three boats in the boatyard have each been repainted completely four times. Each time, the painter covered the entire boat with a single color without scraping off the older paint layers underneath.

The boatyard logbook shows:

- The red paint layer covers the yellow paint layer.
- The green paint layer covers the red paint layer.
- The yellow paint layer covers the white paint layer.

On each boat, the workers replaced one plank at some point between two repaintings. When first fitted onto the boat, the replacement plank was bare, unpainted wood. From the next repainting onward, the new plank was painted along with the rest of the boat, and all paint layers from then on remain intact on the plank:



Paint layers on the replacement planks

A loose slip of paper reads: "Plank replaced after yellow, before red."

Questions:

Which of the three boats—A, B, or C—does that slip belong to? Were the planks on the other two boats replaced earlier or later than this plank?

Hint: A new plank carries only the paint layers applied after it was fitted onto the boat.

Answer: The logbook gives the order **white** → **yellow** → **red** → **green**. A new plank receives only the paint layers applied after it is fitted:

Boat	Paint layers present	When plank was replaced
B	Yellow, red, green	After white, before yellow: earlier than A
A	Red, green	After yellow, before red: matches the slip
C	Green	After red, before green: later than A

Exercise 2: Moving Three Rollers ★★

The workers must move a boat out to the yard. It is lashed to a flat wooden platform 4 m long, whose balance mark (for boat and platform together) is at the middle, 2 m from each end. At first the platform covers the ground from 0 m to 4 m and rests on three rollers centered at A = 0.5 m, B = 2 m, and C = 3.5 m.

The rules:

- The workers push the platform very slowly to the right. The rollers roll without slipping on the ground or under the platform: each time the platform moves 1 m right, the center of each roller under it moves 0.5 m right.[3]
- At every moment (even while stopped to move a roller), at least one roller must support the platform ahead of the balance mark and at least one behind it. A roller exactly under the mark does not count as ahead.
- When a roller reaches the back edge of the platform, the workers may stop, pull it out, carry it to the front, and place its center right under the front edge. They cannot lift the platform or add other supports.

Questions:

To push the platform 4 m to the right in total, at least how many times must the workers move a roller from back to front? Describe how, and check that the platform stays balanced while a roller is moved.

Initial positions from the left end of the platform:
A: 0.5 m; B: 2 m; C: 3.5 m.

Red mark: balance mark at the middle of the platform.

Hint: Because the rollers move forward more slowly than the platform, relative to the platform they gradually shift toward its back edge.

[3] For the motion of an object on rollers, see the Stonehenge problem on NRICH: https://nrich.maths.org/problems/stonehenge

Answer: At least **1 move** is needed. Push the platform 1 m, stop to move A from the back edge to the front edge, and then push another 3 m. All positions below are in meters from the 0 m mark on the ground:

Stage	**Platform**	**Midpoint**	**A**	**B**	**C**
Start	0–4	2	0.5	2	3.5
Push 1 m	1–5	3	1	2.5	4
Move A	1–5	3	5	2.5	4
Push 3 m more	4–8	6	6.5	4	5.5

When A is pulled out, B at 2.5 m and C at 4 m still support both sides of the balance mark at 3 m. Before the move, A is always behind and C is ahead of the balance mark; after the move, B is always behind and A is ahead. At the end, B just touches the back edge of the platform; C at 5.5 m and A at 6.5 m lie on either side of the balance mark at 6 m.

Red mark: balance point. Ground coordinates (m).

Without moving any roller, once the platform moves 3 m, the balance mark is at 5 m and the front roller C is also at $3.5 + 1.5 = 5$ m. No roller remains ahead of the balance mark, so the platform cannot safely travel the full 4 m. Thus, one move is the minimum.

The roads go on

Mira, Bao, Tavi and Suri have only just set out. Volume One follows them through twenty-five places on the Great Map, each with its own story, main puzzle, full solution and two exercises.

The book is given by Math Club & Competitions to its award winners, in English or in Vietnamese: the same stories, pictures and map in both.